Neural Memories

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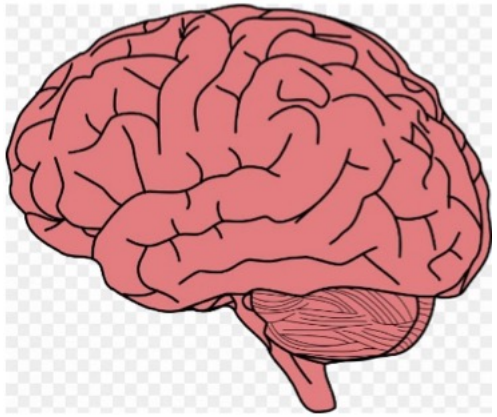
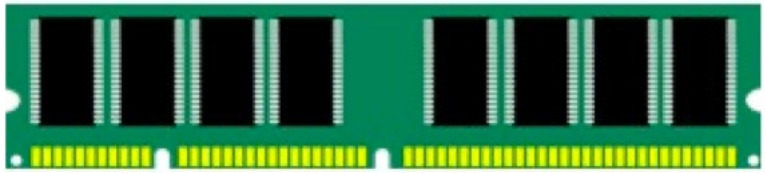
2021/7/16

Mainly borrow from https://neuralreasoning.github.io/slides/Lecture_3_10.pdf

Memory is part of intelligence

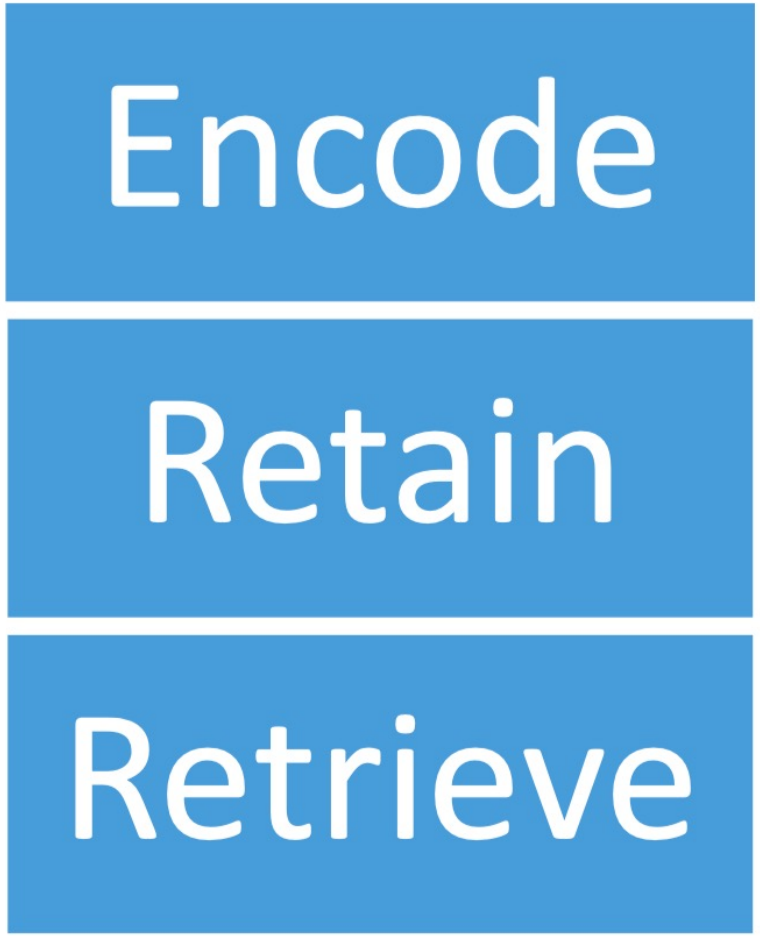
- Memory is the ability to store, retain and recall information
- Brain memory stores items, events and high-level structures
- Computer memory stores data and temporary variables





Common memory activities

- Encode: write information to the memory, often requiring compression capability
- Retain: keep the information overtime. This is often assumed in machinery memory
- Retrieve: read information from the memory to solve the task at hand



Memory taxonomy based on memory content

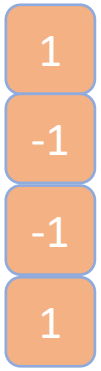
Item Memory	• Objects, events, items, variables, entities
Relational Memory	• Relationships, structures, graphs
Program Memory	• Programs, functions, procedures, how-to knowledge

Item Memory

Associative memory

Hopfield Network

- Associate memory can be implemented as Hopfield network
- Each neuron has integer value
- Neurons form a vector, representing item



Hopfield Network

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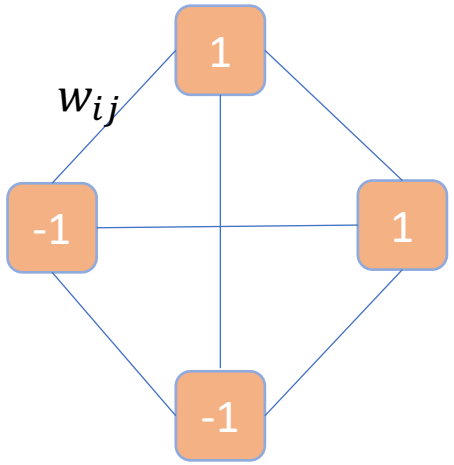
- Connection weights form memory
- Energy function (Neurons' values, weight)



$$E = -\frac{1}{2}\sum_{i,j} w_{ij} s_i s_j + \sum_i \theta_i s_i$$

Hopfield Network

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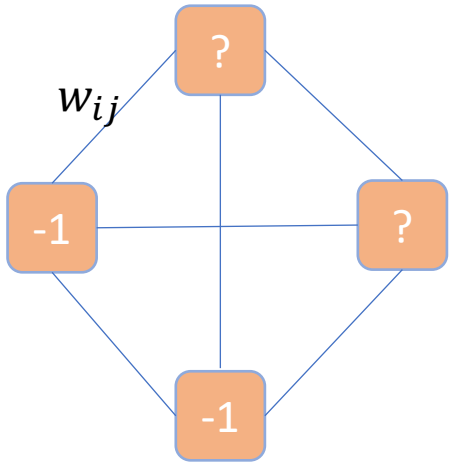
- Energy function (Neurons' values, weight)
- Input a part vector, recovers the rest



- $V_i^{s'} \rightarrow 1$ if $\sum_j w_{ij} V_j^{s'} > U_i$
- $V_i^{s'} \rightarrow -1$ if $\sum_j w_{ij} V_j^{s'} < U_i$

Hopfield Network

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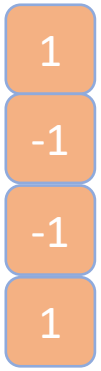

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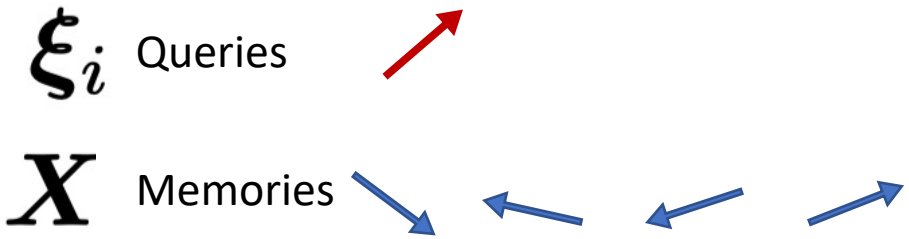
Hopfield network is all you need

- Continuous Hopfield networks retrieve vectors



Hubert Ramsauer et al. Hopfield Network is all you need. ICLR 2021.

Hopfield network is all you need

- Continuous Hopfield networks retrieve vectors

$\boldsymbol{\xi}_i$ Queries

$\boldsymbol{X}$ Memories

Hopfield Energy		New Energy		Update Rule		Transformer
$-\exp\left(\mathrm{lse}\left(1, \boldsymbol{\xi}\boldsymbol{X}^T\right)\right)$	=	$-\mathrm{lse}(\beta, \boldsymbol{\xi}\boldsymbol{X}^T) + \frac{1}{2}\boldsymbol{\xi}^T\boldsymbol{\xi} + c$	⇨	$\mathrm{softmax}\left(\beta\, \boldsymbol{\xi}\boldsymbol{X}^T\right) \boldsymbol{X}$	=	$\mathrm{softmax}\left(\frac{1}{\sqrt{d_k}}\, \boldsymbol{Q}\boldsymbol{K}^T\right) \boldsymbol{V}$

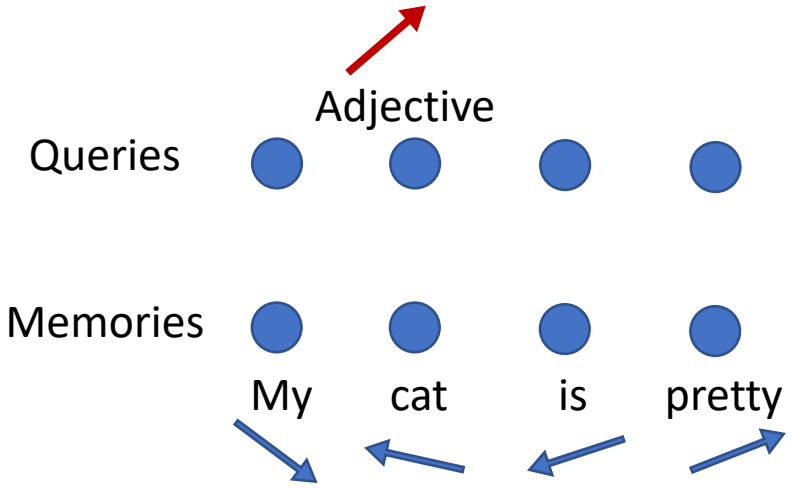
$$\mathrm{lse}(\beta, \boldsymbol{x}) \;=\; \beta^{-1} \log\left(\sum_{i=1}^{N} \exp(\beta x_i)\right) \;,$$

Converge to minimum

Hubert Ramsauer et al. Hopfield Network is all you need. ICLR 2021.

Hopfield network is all you need

- Continuous Hopfield networks retrieve vectors



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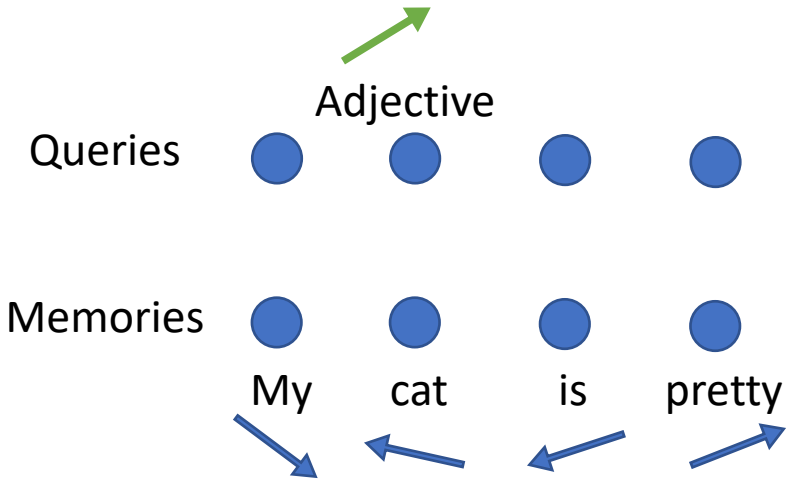
Hopfield network is all you need

- Continuous Hopfield networks retrieve vectors
  - Repeat the updating rule multiple times: retrieve the key vector



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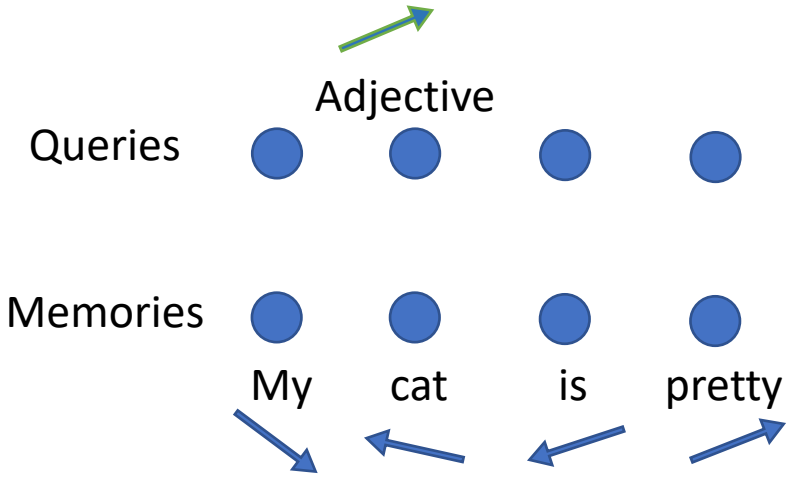
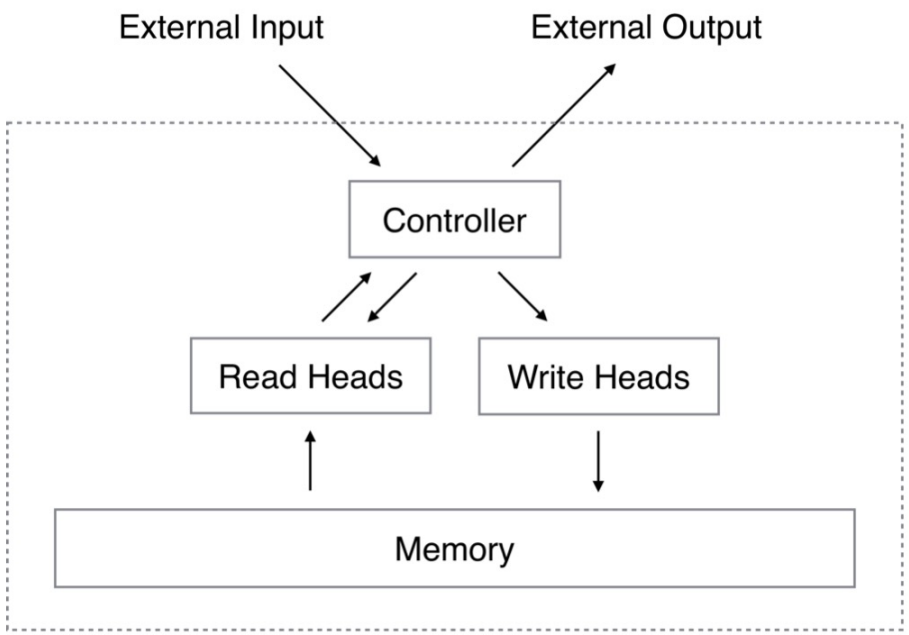


Hubert Ramsauer et al. Hopfield Network is all you need. ICLR 2021.

Neural Turing Machine (NTM)

- Memory is a 2d matrix
- Controller is a neural network
- The controller read/writes to memory at certain addresses.
- Simulate Turing Machine
- Support symbolic reasoning, algorithm solving

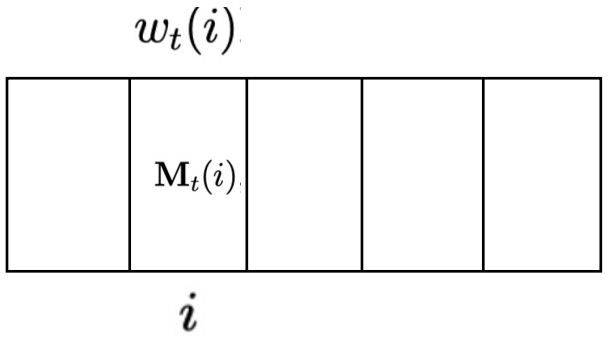


Graves, Alex, Greg Wayne, and Ivo Danihelka. "Neural turing machines." arXiv preprint arXiv:1410.5401 (2014).

Neural Turing Machine (NTM)

- Reading

$$\mathbf{r}_t \longleftarrow \sum_i w_t(i)\mathbf{M}_t(i),$$

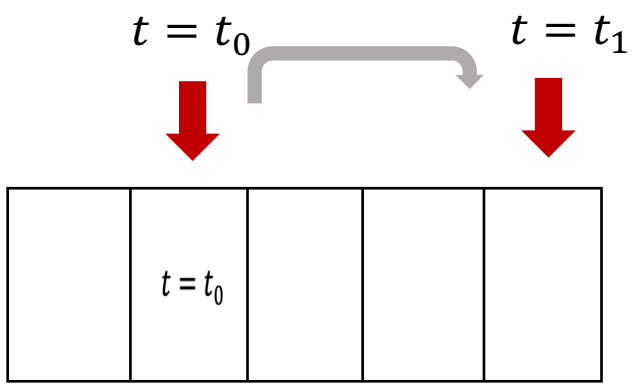


- Writing
  - Erase $$\tilde{\mathbf{M}}_t(i) \longleftarrow \mathbf{M}_{t-1}(i)\left[\mathbf{1} - w_t(i)\mathbf{e}_t\right],$$
  - Add $$\mathbf{M}_t(i) \longleftarrow \tilde{\mathbf{M}}_t(i) + w_t(i)\,\mathbf{a}_t.$$

Graves, Alex, Greg Wayne, and Ivo Danihelka. "Neural turing machines." arXiv preprint arXiv:1410.5401 (2014).

Neural Turing Machine (NTM)

- Addressing by weight shifting
  1. Focusing on content
     - similarity between key vector $\mathrm{k_t}$ and memories



$$w_t^c(i) \longleftarrow \frac{\exp\left(\beta_t K\left[\mathbf{k}_t, \mathbf{M}_t(i)\right]\right)}{\sum_j \exp\left(\beta_t K\left[\mathbf{k}_t, \mathbf{M}_t(j)\right]\right)}. \qquad K\left[\mathbf{u}, \mathbf{v}\right] = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}|| \cdot ||\mathbf{v}||}.$$

  2. Focusing on location: circular convolution

$$\tilde{w}_t(i) \longleftarrow \sum_{j=0}^{N-1} w_t^g(j)\, s_t(i-j)$$

Graves, Alex, Greg Wayne, and Ivo Danihelka. "Neural turing machines." arXiv preprint arXiv:1410.5401 (2014).

Neural Turing Machine (NTM)

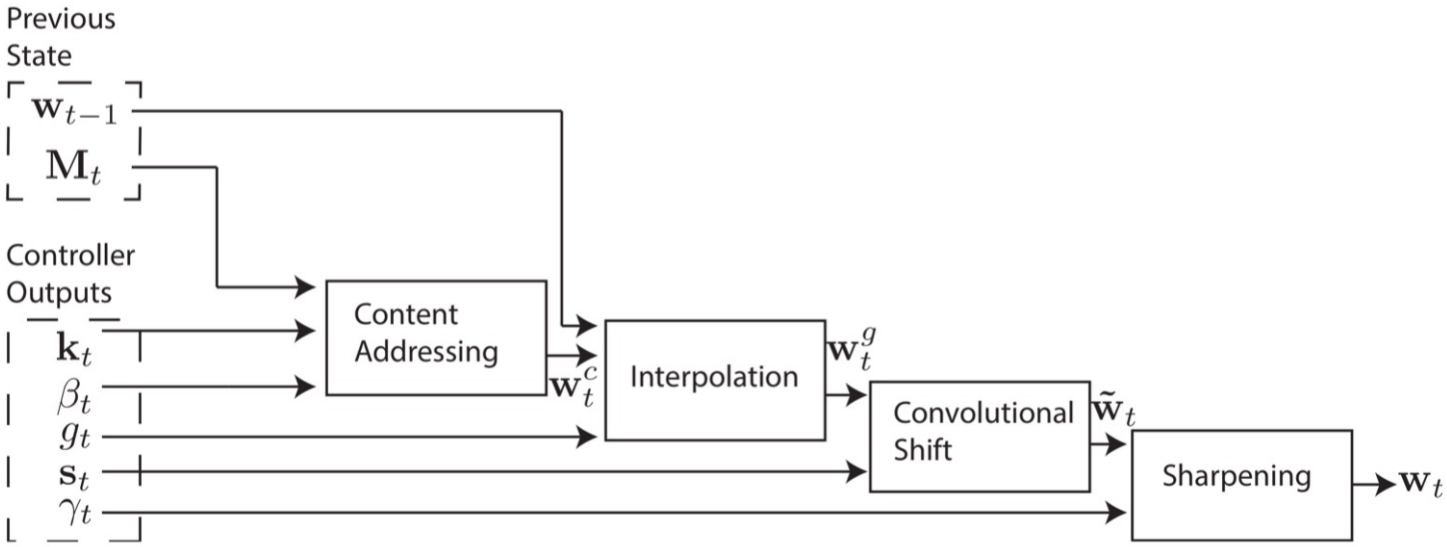
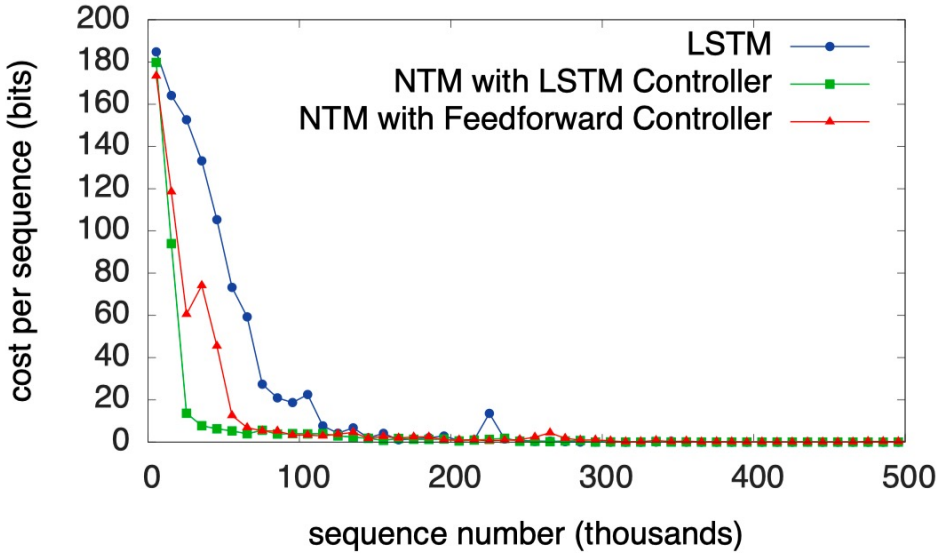


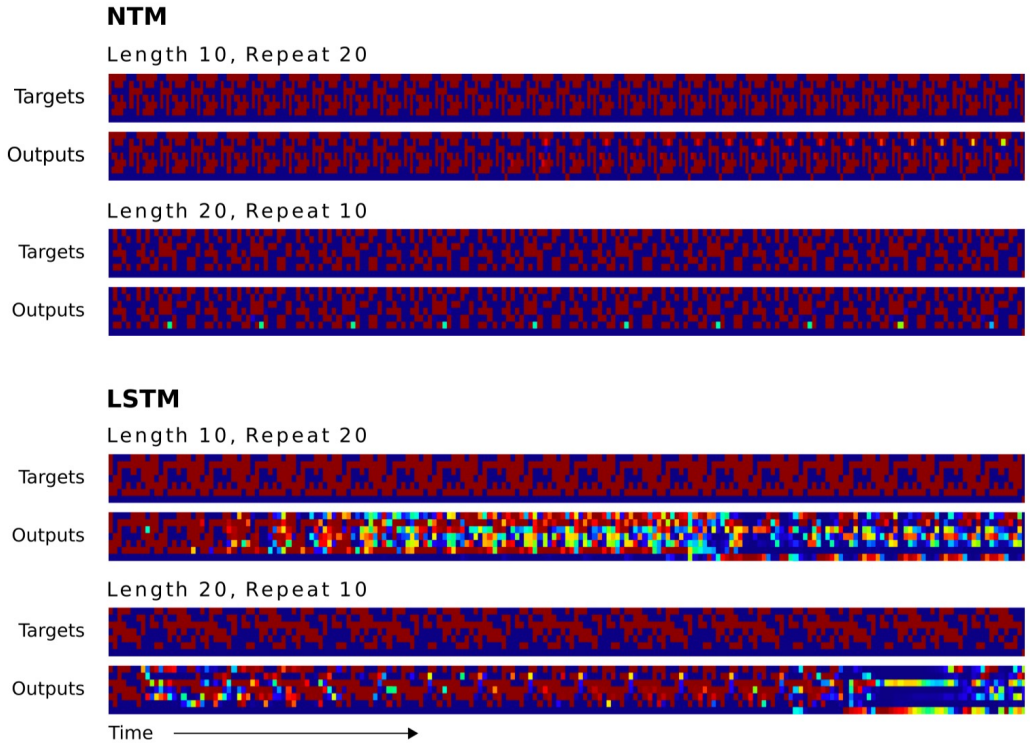
Figure 2: Flow Diagram of the Addressing Mechanism. The *key vector*, $\mathbf{k}_t$, and *key strength*, β_t, are used to perform content-based addressing of the memory matrix, $\mathbf{M}_t$. The resulting content-based weighting is interpolated with the weighting from the previous time step based on the value of the *interpolation gate*, g_t. The *shift weighting*, $\mathbf{s}_t$, determines whether and by how much the weighting is rotated. Finally, depending on γ_t, the weighting is sharpened and used for memory access.

Graves, Alex, Greg Wayne, and Ivo Danihelka. "Neural turing machines." arXiv preprint arXiv:1410.5401 (2014).

Neural Turing Machine (NTM)

- Experiments:
  - Copy, Repeat copy, Priority sort, etc.





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Relational Memory

Graph memory

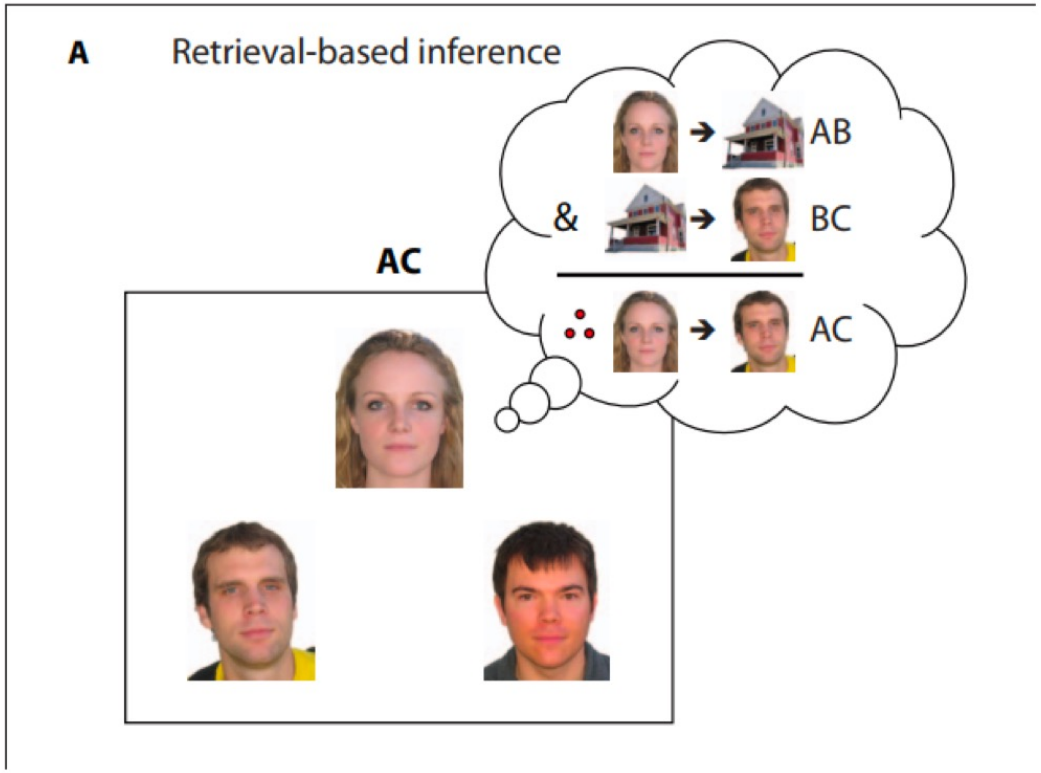
Tensor memory

Motivation for relational memory: item memory is weak at recognizing relationships

Item Memory

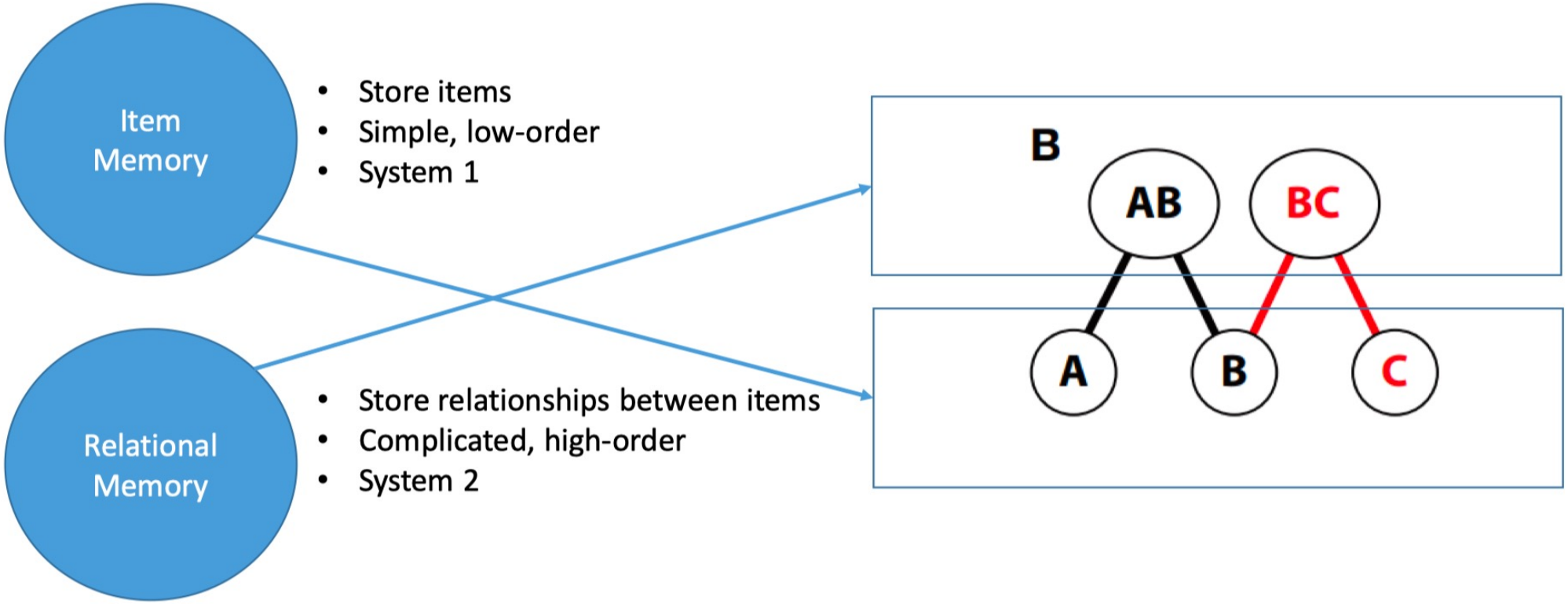
- Store and retrieve individual items
- Relate pair of items of the same time step
- Fail to relate temporally distant items

$$\hat{\mathbf{M}} = \sum_{k=1}^{q} \mathbf{b}_k \mathbf{a}_k^T$$



Howard Eichenbaum, Memory, amnesia, and the hippocampal system (MIT press, 1993).
Alex Konkel and Neal J Cohen, "Relational memory and the hippocampus: representations and methods", Frontiers in neuroscience 3 (2009).

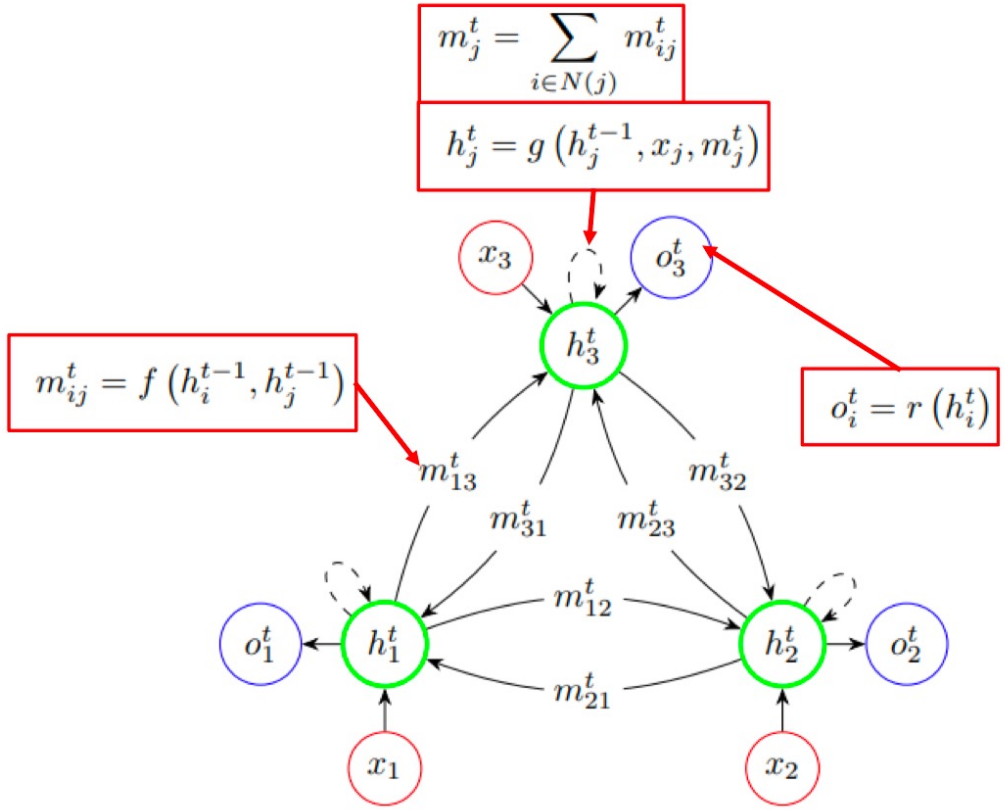
Dual process in memory



Howard Eichenbaum, Memory, amnesia, and the hippocampal system (MIT press, 1993).
Alex Konkel and Neal J Cohen, "Relational memory and the hippocampus: representations and methods", Frontiers in neuroscience 3 (2009).

Memory as graph

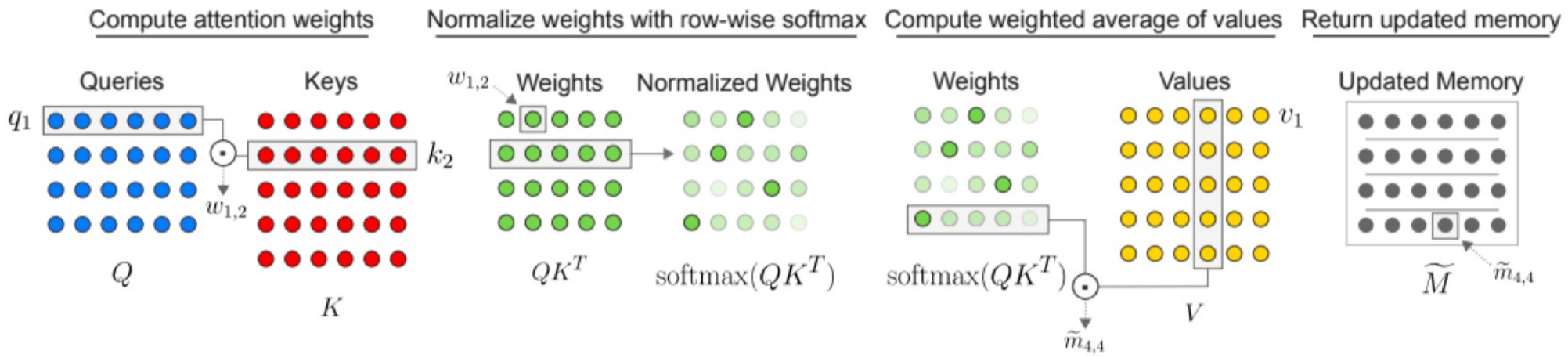
- Memory is a static graph with fixed nodes and edges
- Relationship is somehow known
- Each memory node stores the state of the graph's node
- Write to node via message passing
- Read from node via MLP



Palm, Rasmus Berg, Ulrich Paquet, and Ole Winther. "Recurrent Relational Networks." In NeurIPS. 2018.

Capturing relationship can be done via memory slot interactions using attention

- Graph memory needs explicit design of nodes and edges
- Can we automatically learns structure with a 2d tensor memory?
- Capture relationship: each slot interacts with all other slots (self-attention)



Santoro, Adam, Ryan Faulkner, David Raposo, Jack Rae, Mike Chrzanowski, Théophane Weber, Daan Wierstra, Oriol Vinyals, Razvan Pascanu, and Timothy Lillicrap. "Relational recurrent neural networks." NeurIPS. 2018.

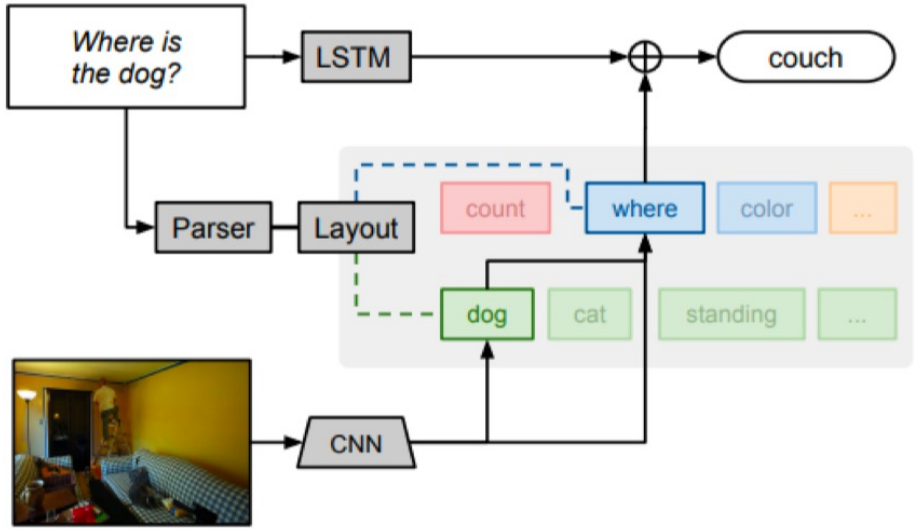
Program Memory

Module memory

Stored-program memory

Module Memory – Neural Module Networks

- Remember how to execute a program by remembering sub-modules (seen as a program)
- A program designed for a task becomes a module



Andreas, Jacob, Marcus Rohrbach, Trevor Darrell, and Dan Klein. "Neural module networks." CNVP2016.

Neural Stored-Program Memory (NSM)

- NTM stores only the data, the controller is a fixed NN, one for each task.
- NSM stores the programs, i.e. the controller.

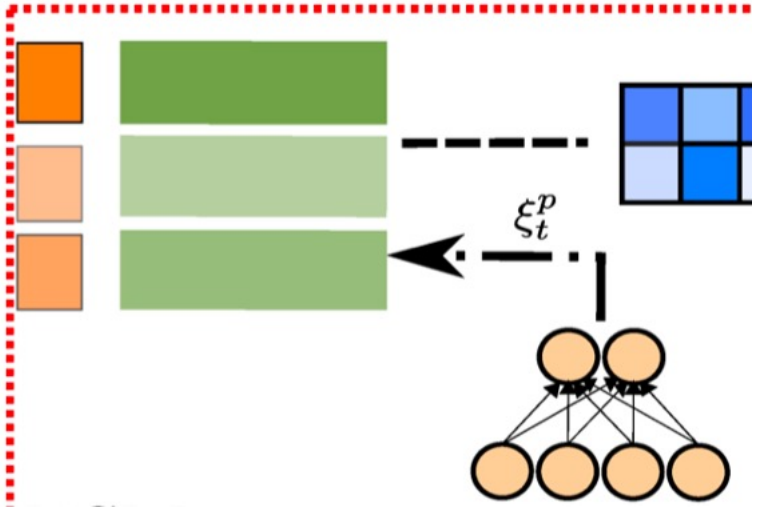
Le, Hung, Truyen Tran, and Svetha Venkatesh. "Neural Stored-program Memory." In International Conference on Learning Representations. 2019.

Neural Stored-Program Memory (NSM)

- NTM stores only the data, the controller is a fixed NN, one for each task.
- NSM stores the programs, i.e. the controller.
- A NSM is a key value memory $\mathbf{M}_p \in \mathbb{R}^{P \times (K+S)}$
- Memory about a program has two parts:

$$\mathbf{M}_p(i).k \text{ and } \mathbf{M}_p(i).v$$

- Use a vector ξ_t^p to query the program (v) by key (k)
- Use a interface network to generate ξ_t^p

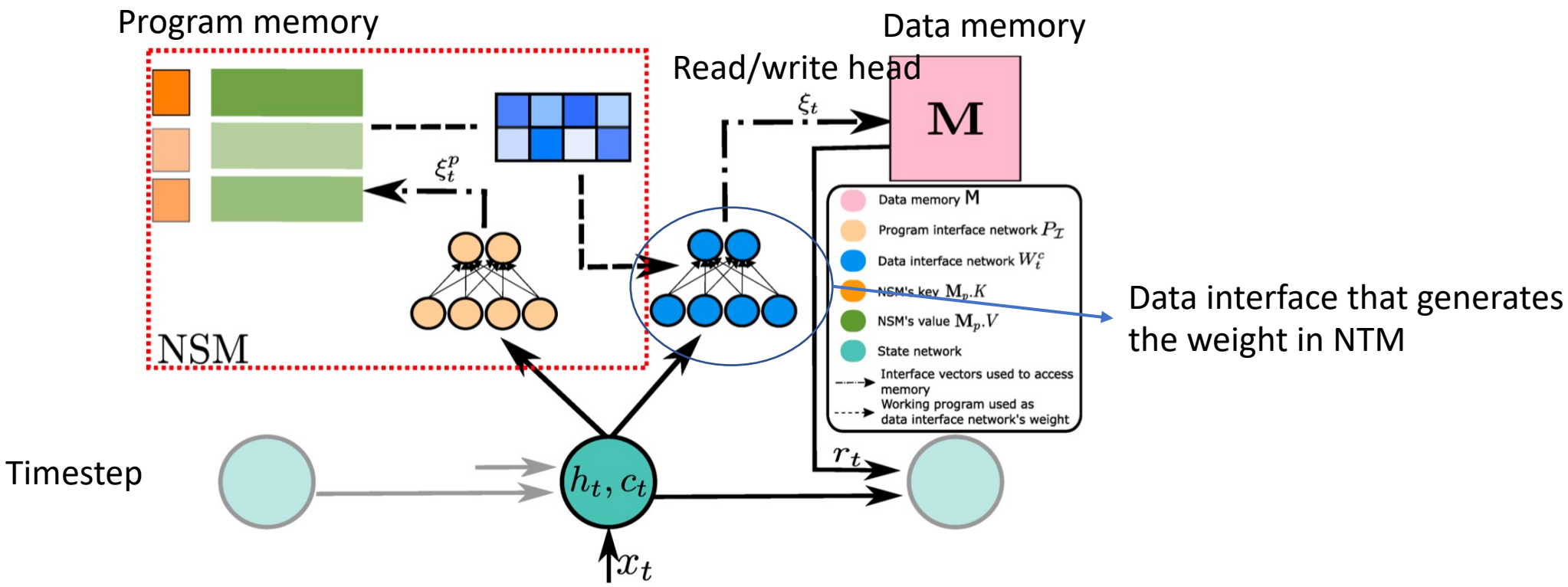


Interface Network

Le, Hung, Truyen Tran, and Svetha Venkatesh. "Neural Stored-program Memory." In International Conference on Learning Representations. 2019.

Neural Stored-Program Memory (NSM)

- A NSM is a key value memory



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Neural Stored-Program Memory (NSM)

- Experiments
  - Superior even in single tasks (Set P=2, believe that the task can be divided into 2 sub tasks)

Task	Copy	R. Copy	A. Recall	D. N-grams	P. Sort	L. Copy
NTM	**0.00**	405.10	7.66	132.59	24.41	16.04
NUTM (p=2)	**0.00**	**366.69**	**1.35**	**127.68**	**20.00**	**0.02**

  - Sequencing Tasks
    - A dysfunctional brain may have difficulty in changing from one task to the next and get stuck in its preferred task (perseveration phenomenon).

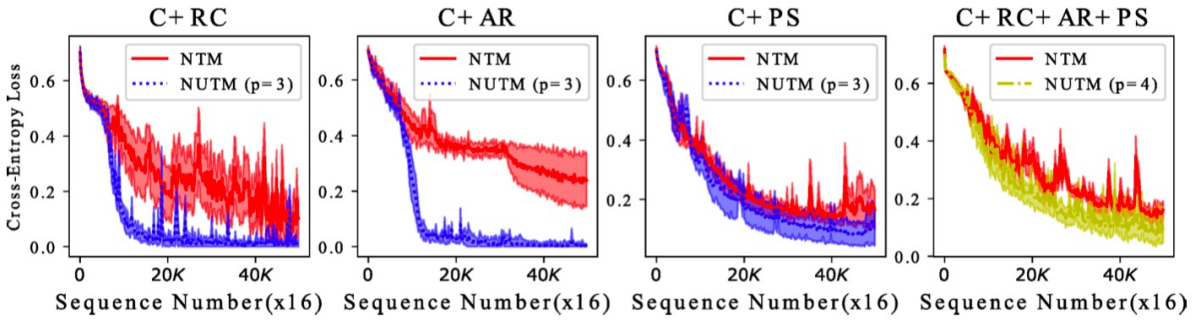
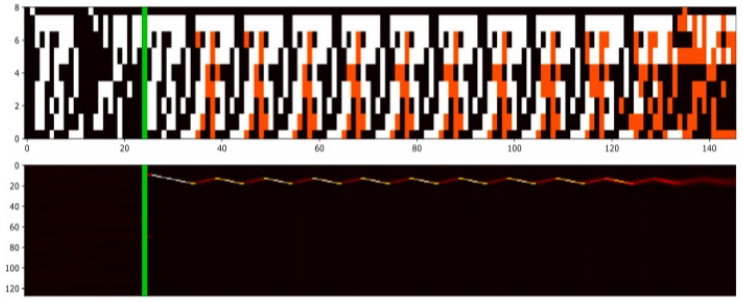


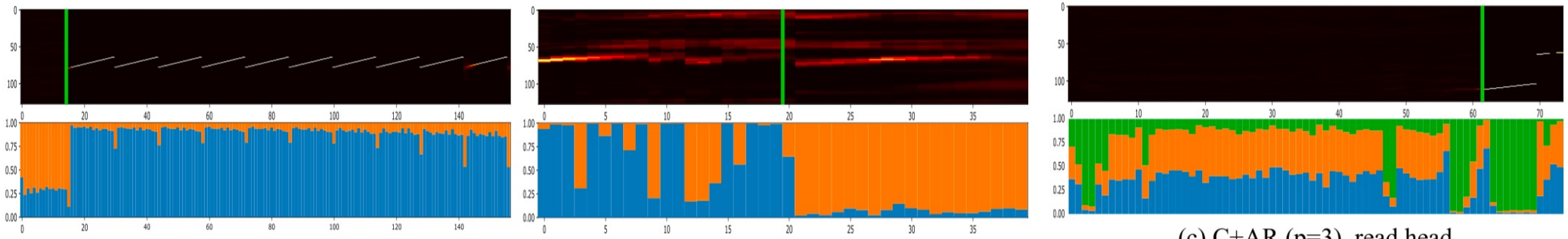
Figure 4: Learning curves on sequencing NTM tasks.



(d) Perseveration in NTM (C+RC), read head

Neural Stored-Program Memory (NSM)

- Experiments
  - Visualization of program selection.



(a) Repeat copy (p=2), read head (b) Priority sort (p=2), write head (c) C+AR (p=3), read head

  - Can also adapted to other data-memory networks.
  - Can improve few-shot learning

Le, Hung, Truyen Tran, and Svetha Venkatesh. "Neural Stored-program Memory." In International Conference on Learning Representations. 2019.

Thank you for listening